Exercise 2 (Marching Cubes)

The Marching Cubes algorithm is a computer graphics method used to extract a 3D surface mesh (made of triangles) from a 3D grid of scalar data.

In this exercise we will identify which case we obtain after applying the Marching Cubes algortihm for the isovalue $c=0$ (the threshold) by given the cells from Figure 2.

Considering the indexing of the vertices given in the slides:

For cell 1:

Vertices 3 and 8 have negative values. The index is 11011110. We are therefore in case 4, and we have to disambiguate the upper face as it is the only one to have different signs at its diagonals.

By substituting in the asymptoting decider, we get $(-2\cdot-2-1\cdot1)/(-2-1-1-2)=-1/2<0$ so the asymptotic decider tells us that the center of the face should have a negative value and thus that we should have two positive regions. Therefore there should be a tunnel connecting edges 3 and 8.

For cell 2:

Again, vertices 3 and 8 have negative values. The index is 11011110. We are again in case 4 and thus we have to use the asymptotic decider to disambiguate the upper face.

In this case, we get $(-1\cdot-1-2\cdot2)/(-1-2-2-1)=1/2 > 0$, so the asymptotic decider tell us that the intersection of the asymptotes should have a positive value, and thus that there should be two negative regions around corners 3 and 8. We are therefore in case 4 as it is shown in the sheet.

For cell 3:

Now vertices 1, 3, 6 and 8 have negative values. The index is 01011010. We are therefore in case 8, and we have to disambiguate all 6 faces.

As all faces are the same as the ambiguous face of cell 1, we see that the center of all faces should be negative and the positive corners should be isolated. Therefore, the result should look as case 8 in the sheet, with the positive corners at the marked points.